Contract No. DD-2023-118

Construction Contract

for

Diriyah Gate II Infrastructure Package 1

between

Diriyah Gate Company Limited

as the Employer

and

Al-Ayuni Investment & Contracting Company

as the Contractor

Volume 2 of 6

Specification (5 of 9)

DGCL Environment Requirements for Contract &

DGCL Sustainability Requirements

TABLE OF CONTENTS

VOLUME 2 - SPECIFICATION

CONTENTS

"For the purpose of Sub-Clause 1.5.1(d) of the Conditions of Contract, the order of precedence of documents within the Specification shall be set out as follows"

- Post Tender Clarifications
- Tender Addenda
- Schedule of Project Requirements
- Particular Specification
- **DGCL Environment Requirements for Contract & DGCL Sustainability Requirements**
- DGCL HSE Specification
- DGCL BIM & Digital Delivery Pack
- DGCL General Specification
- Miscellaneous

VOLUME 2 – SPECIFICATION

DGCL Environment Requirements for Contract

Project Name:	Diriyah Gate Development
Works Package:	All
Document:	**Environment Requirements**
Revision Date:	23 January 2023

CONTENTS

1. ENVIRONMENT REQUIREMENTS FOR CONTRACTORS

1.1 GENERAL REQUIREMENTS

The scope of works as defined by the tender documents shall, in addition to whatever is described in there, include responsibility for the management and control of all Environmental aspects of the project and the site.

Such responsibility shall not be limited to the project but shall extend to the works and activities of all appointed Contractors, including nominated subcontractors, suppliers, and other persons working on or visiting the site.

The Employer's development guidelines reflect the sustainable design, construction, and operation requirements to achieve the following sustainability certifications, as required:

- LEED for Cities & Communities Gold certification (version 4.1), with aspirational target of Platinum certification
- Mostadam for Communities Gold certification (version 2019), with aspirational target of Platinum certification
- SITES Rating System for Sustainable Land Design and Development (version 2)
- LEED Building Design and Construction (version 4), certification applicable for specific assets
- Mostadam for Residential Buildings Design and Construction (version 2019), certification applicable for specific assets
- Mostadam for Commercial Buildings Design and Construction (version 2019), certification applicable for specific assets
- Parksmart Gold certification for parking structures

The Contractor is responsible for implementing all requirements during construction to meet and achieve the project's sustainability objectives and targets. Refer to the Sustainability Scope of Service for Contractors for additional details.

It is the Contractor's responsibility to ensure compliance with the statutory environmental requirements of the Kingdom of Saudi Arabia (KSA), as well as other applicable competent authorities. The competent authority for issuance of environmental approvals and permits for projects, as well as specific construction activities the National Center for Environmental Compliance (NCEC). Additional legal requirements may be detailed in approvals or permits issued by other applicable competent authorities including, but not limited to, the following:

- Ministry of Environment, Water and Agriculture (MEWA)
- National Center for Waste Management (MWAN)
- The Ministry of Industry and Mineral Resources (MIM)
- The Saudi Commission for Tourism and National Heritage (SCTH)
- The National Center for Wildlife
- The Royal commission for Riyadh City (RCRC)
- Saudi Geological Survey (SGS)
- Employer's Urban Planning and Municipal Affairs Department (UMA)

The contractor shall be responsible for payment of commercial penalties, and implementation of corrective and preventative action resulting from violations issued by applicable competent authorities. Authorities include, but are not limited to the NCEC, and UMA. Refer to *Appendix A* "Article (11) – Controlling Violations and Imposing Penalties" for table of penalties enforceable for non-compliance. This is extracted from MEWA's Implementing Regulation of Environmental Permits for Establishing and Operating Business Activities, Environment Law Issued by Royal Decree No. (M/165), dated 19/11/1441 A.H.

1.2 FAILURE TO COMPLY

Should the Contractor fail to comply with any of the requirements of this document, the Engineer shall notify the Contractor in writing of this situation. Upon receiving such notification, the Contractor shall immediately take all necessary corrective and preventative actions. Any corrective and preventative action, unless provided otherwise in this contract, be taken at the Contractor's expense. If the Contractor fails to take prompt corrective and preventative action, the Engineer may;

(a) Direct the Contractor to suspend all or part of the works until satisfactory corrective and preventative action has been implemented;
(b) Perform the corrective and preventative actions on its own, with all costs back-charged to the Contractor's account; or
(c) Terminate the contract. Costs incurred by the Contractor as a result of such work suspension or termination shall be solely the Contractor's responsibility, and in the case of suspension, any resultant delays shall not be deemed excusable hereunder.

1.3 PRE-CONSTRUCTION SURVEY REQUIREMENTS

Prior to commencement of construction activities, the following shall be completed:

1. The Contractor shall attend the environment and sustainability induction, after which reference documentation shall be provided for the Contractor to fulfil its obligations.
2. The Contractor shall conduct a pre-construction survey of the site, and submit to the Engineer for review and approval. The survey shall include baseline monitoring and photographic record, to determine the air quality (PM10 and PM 2.5), noise, and vibration conditions at the project boundary, sensitive receptor locations, as well as items of ecological importance and identified heritage assets.

1.4 CONSTRUCTION ENVIRONMENT AND SUSTAINABILITY MANAGEMENT PLAN (CESMP) REQUIREMENTS

Based on the Employer's construction activities, associated environmental aspects and potential impacts, each project shall prepare a project-specific Construction Environment and Sustainability Management Plan (CESMP) to control activities, monitor aspects and minimise environmental impacts.

The CESMP serves as the primary reference document to ensure compliance with all legal requirements, deliverables, implementation of environmental best practices, and details the project's Environmental Management System (EMS), which shall be compliant with ISO 14001 (2015) requirements.

The CESMP shall be compiled by the Contractor's suitably qualified Environment Professional and submitted to the Engineer for review and approval, within 28 days of contract award. The CESMP shall be revised by the Contractor periodically to remain applicable to the project's scope, activities and updated regulatory conditions as required. Additional guidance for CESMP development is provided in *Appendix B*.

1.5 ENVIRONMENT MONTHLY REPORTING REQUIREMENTS

The Contractor shall comply with the Employer's environment monthly reporting requirements and submission processes, which may be revised by the Employer as required, to fulfil updated regulatory conditions and / or environmental best practices.

The monthly reporting cycle starts on the 24th of each month and runs up to and including the 23rd day of the following month. For example, information and data to be included in June 2022's Environment Monthly Report (EMR) shall include information and data from the 24th of May 2022, up to and including the 23rd of June 2022.

The EMR shall be signed / endorsed by the Contractor's appointed responsible person (i.e. Project Manager), and submitted to the Engineer for review and approval no later than the 26th day of the reporting month.

Project stakeholders are encouraged to communicate informally (e-mail, discussions) regarding the content of the EMR to ensure accuracy prior to formal submission to the Engineer. Additional guidance for EMR development is provided in *Appendix C*. Refer to the Sustainability Scope of Service for Contractors for sustainability reporting requirements.

1.6 ENVIRONMENT TEAM

The Contractor shall ensure that adequate resources (in terms of manpower, equipment, or finance) are provided for effective implementation of the CESMP throughout the project's construction phase.

Contractors shall appoint at least one full time, site-based Environment Manager that will be the focal point between the Contractor and project entities for all environmental matters. The number and seniority of required Environment Professionals shall be aligned with project-specific deliverables, project scale, and complexity to ensure compliance and delivery of the project's objectives and targets.

Contractors shall ensure that the CV of the proposed Environment Professional is submitted to the Engineer for review and approval prior to candidate hiring.

As part of monitoring and review by the Engineer, if it is observed that Environment Professionals are not competent to perform their duties, a recommendation will be made to replace said staff. Contractors shall provide the Engineer with advanced notification in writing in case of any replacement/resignation of the Environment Professionals.

Table 1.6.1 below should be used as a guideline only to determine the specific title and seniority of the Contractor's Environment Professionals.

Table 1.6.1: Guideline Criteria for Contractor's Environment Professionals

Title	Total Experience	Experience in similar Role (Project Scale / Complexity)	Required Qualifications	Preferred Qualifications / Experience
Environment Manager	More than 10 Years	8 Years	BSc. In Environmental Science / Engineering or similar	Postgraduate qualification in related discipline
				Certification with a recognised professional body.
				More than 5 Years GCC Experience
			GCC Experience	LEED AP and/or Mostadam AP, and SITES AP
Environment Senior Officer	7 Years	5 Years	BSc. In Environmental Science / Engineering or similar	Postgraduate qualification in related discipline
			GCC Experience	Certification with Sustainability Ratings Scheme (LEED AP and/or Mostadam AP, and SITES AP)
Environment Officer	4 Years	2 Years	BSc. In Environmental Science / Engineering or similar	GCC Experience.
				Certification with Sustainability Ratings Scheme (LEED AP and/or Mostadam AP, and SITES AP)

Table 1.6.2 below provides the general environmental management responsibilities related to the Contractor's key personnel.

Table 1.6.2: Responsibilities for Contractor's Key Personnel

Position/ Role	Main Environmental Management Responsibilities
Project Director/ Manager	○ Approve the CESMP ○ Ensure the required resources for the development and implementation of the CESMP ○ Facilitate and provide necessary support for communication related to environmental issues raised by interested parties

Position/ Role	Main Environmental Management Responsibilities
	○ Require from subcontractors to adhere to all environmental requirements applicable to their activities ○ Chair the environmental management review
Procurement Manager	○ Ensure that environmental requirements are considered when purchasing materials and selecting subcontractors and service providers
HSE Manager	○ Support the Environment Manager on the development and implementation of the CESMP, namely concerning the applicable integrated Management Systems procedures ○ Coordinate with the Environment Manager for inductions ○ Conduct internal audits
Environment Manager	○ Act as the principal point of advice in relation to all matters concerning the environmental performance of the project ○ Ensure environmental risks of the project are identified and appropriate mitigation measures are established ○ Prepare, review, implement, and update the CESMP ○ Prepare and follow the processes to obtain all the required environmental permits ○ Supervise the implementation of the CESMP ○ Manage the Senior Environment Officer, and Environment Officer ○ Coordinate and chair environmental meetings with project stakeholders, and third parties as required ○ Attend relevant project meetings, including progress meetings. Arrange meetings with the construction team to ensure that variations on scope of the work will be addressed and impacts to environment will be mitigated ○ Identify training needs, coordinate and conduct environmental training ○ Conduct and coordinate weekly site inspections at work sites ○ Conduct and coordinate monthly site inspections at work sites ○ Attend audits and inspections carried out by Authorities, Client, PMC, CSC, Environmental Monitoring Consultant, and others as required ○ Ensure that environmental monitoring is carried out, in accordance with the CESMP requirements and compile environmental monitoring records ○ Coordinate and conduct internal environmental audits with the Senior Environment Officer and/or Environment Officer support ○ Establish corrective and preventive actions and evaluate its effectiveness

Position/ Role	Main Environmental Management Responsibilities
	○ Receive environmental incident reports, investigate and establish recovery, corrective, and preventative actions ○ Ensure that required environmental records are maintained, and accurate ○ Ensure that data and evidences for the contractor monthly environment report/ logs are collected and recorded and review the report/ logs ○ Implement environmental awareness campaigns, conduct specific environmental training, compile and share lessons learnt
Environment Senior Officer / Environment Officer	○ Ensure that the environmental safeguards and management measures in the CESMP are being implemented across the site ○ Assist the Environment Manager in the preparation of documentation, and in the implementation of the CESMP ○ Undertake environmental toolbox talks and inductions ○ Conduct daily site walkover, monitor and record site compliance including air quality and noise conditions ○ Conduct on-site monitoring ○ Report all environmental issues, observations, and non-conformances to the Environment Manager ○ Monitor workers performance (including subcontractors) on site with regards to Environment ○ Communicate environmental instructions or information (from the CESMP or the Environmental Manager) to staff on site ○ Collaborate on the collection of data and evidences for the Contractor's monthly environment report/ logs
Construction Manager	○ Ensure onsite compliance of the CESMP ○ Coordinate with Project Manager and Environment Manager for the environmental training need identification ○ Ensure that any environmental incidents are reported at the earliest possible and emergency procedures initiated ○ Control and monitor actions required by the CESMP ○ Monitor sub-contractor performance and commitment ○ Communicate with the environment team on all aspects concerning environmental issues and report to the Environment Manager as required ○ Liaise with the Environment Manager to propose corrective and preventative actions, and ensure implementation of the agreed corrective and preventative actions

Position/ Role	Main Environmental Management Responsibilities
	o Ensure collaboration of the site team on the collection of data and evidences for the Contractor's monthly environment report/ logs

1.7 ENVIRONMENT AND SUSTAINABILITY (E&S) KEY PERFORMANCE INDICATORS (KPIs)

The E&S KPIs reflect the Employer's minimum compliance requirements, and serve as a tool to identify non-compliances, areas for improvement, appropriate mitigation measures, as well as good E&S practices on the project site. Contractor performance shall be assessed monthly, and at completion of construction activities as detailed in the following sections. The Employer shall maintain a record of the project's performance, which will inform the assessment and award of future construction contracts.

1.7.1 MONTHLY E&S KPIs SCORING

E&S KPIs shall be compiled by the Engineer on a monthly basis using the below template Table 1.7.1. For each of the ten individual KPIs, the Engineer will enter the corresponding score (either 1 or 0) into the applicable cell. The total monthly score is then added and converted to a percentage to determine the level of compliance achieved by the Contractor for a specific month, which shall be captured in the Engineer's monthly project progress report issued to the Employer.

Table 1.7.1: Monthly KPIs Scoring Template

KPI #	Description	ENGINEER	
		Score (1/0)	Comments
1	A project specific CESMP has been approved by the Engineer		
2	A suitably qualified, full-time Environment and Sustainability Professional is appointed on site		
3	Air quality, noise and vibration monitoring is conducted in compliance with CESMP and Project Environmental Permit		
4	Appropriate mitigation are measures implemented to reduce noise and vibration exceedances, where applicable		
5	Appropriate mitigation measures are implemented to reduce air quality exceedances, where applicable		
6	Monthly Environment and Sustainability Report is submitted within required time frame		
7	Monthly Environment and Sustainability Report includes all required information, data reported is accurate and verifiable		
8	Site inspections are conducted by the Contractor according to the approved CESMP		
9	Non-conformances and observations are closed in a timely manner (within the agreed time frame)		
10	Energy and water reduction measures are implemented on site per the approved CESMP requirements		
	TOTAL Score		
	TOTAL Score (%)		

Table 1.7.1.1: Monthly E&S KPIs KPIs Compliance Category

Category	Monthly Score Achieved	Description	Guidance
1	100%	Fully Compliant	Identify opportunities for best practices and continual improvement
2	90%	Minor Non-Compliance	Prioritise corrective and preventative action to address non-compliance

3	80%	Non-Compliant	Take immediate corrective and preventative action to address multiple non-compliances
4	Less than 80%	Major Non-Compliance	Take immediate corrective and preventative action to address major project non-compliances

1.7.2 CONSTRUCTION COMPLETION E&S KPIs SCORING

Construction completion E&S KPIs shall be compiled by the Engineer at the construction completion stage, and are assessed individually as detailed in Table 1.7.2 below. Applicability of each KPI shall be determined by the Engineer based on the specific project's scope of work. Applicable KPIs are scored by the Engineer as either compliant (score of 1), or non-compliant (score of 0), and reported to the Employer accordingly.

Table 1.7.2: Construction Completion KPIs Scoring Template

KPI #	**Description**	**ENGINEER**		
		Score (1/0)	**Compliant / Non-Compliant / NA**	**Comments**
11	75% or more (by weight) C&D waste from project diverted from landfills (including evidence)			
12	40 % (by cost) of permanently installed top five infrastructure / public realm materials meet at least one sustainable sourcing and extraction requirement (including evidence)			
13	50% or more aggregates sourced (by weight) are recycled aggregates (including evidence)			
14	30% or more materials (by cost) sourced from within KSA (including evidence)			
15	10% or more materials (by cost) sourced from one or more of the following: reused materials (excluding aggregates), reclaimed timber, recycled rubber, and/or recycled steel (including evidence)			

1.8 PROJECT CLOSE OUT

Upon project completion the Contractor shall ensure that all environment requirements have been met, and copies of all applicable documentation relating to the management and implementation of environment requirements, is submitted to the Engineer for acceptance.

Once the Contractor is satisfied that demobilisation of all temporary work areas, temporary installations, plant and equipment, general and hazardous material / waste storage areas meet environment requirements, the Engineer shall be notified. The Contractor and Engineer shall agree a suitable date and time to conduct the Project Close Out Inspection, to identify and agree on any outstanding items of concern for the Contractor to address. The Project Close Out Inspection Checklist shall be completed by the Contractor and submitted to the Engineer for review and approval, prior to final demobilisation of all personnel from the project.

APPENDIX A

Article (11) – Controlling Violations and Imposing Penalties, extracted from Implementing Regulation of Environmental Permits for Establishing and Operating Business Activities, Environment Law Issued by Royal Decree No. (M/165), dated 19/11/1441 A.H.

Article (11) - Controlling Violations and Imposing Penalties

Violations of the provisions of this Regulation shall be identified and the penalties shown in Table (2) shall be imposed in accordance with the implementing regulation for controlling violations and imposing penalties of the Environment Law, taking into account the following:

(1) The fine for serious violations shall be estimated according to the degree of damage, the inherent value of the damaged site, its area, the type of damaged receptors, and the economic and social impacts of that damage.

(2) The fine for serious violations mentioned in Clause (1) of this Article shall be determined by a committee, including specialists and qualified persons formed by a decision made by CEO of the Center.

(3) The violation shall be serious if any of the following are proven:

A. Actions mentioned in Article 35 of the Law.

B. Actions that lead to an Environmental Degradation.

C. Actions that damage Sensitive Receptors or Environmentally Sensitive Areas.

Table (2) - Violations and Penalties

No.	Violation	Fine/SAR	Notes
1	Providing incorrect information, data, or results in	From 1,000 to 100,000	Correcting the violation and referring the violator to the Public Prosecution.

No.	Violation	Fine/SAR	Notes
	any study, plan, or report submitted to the Center.		
2	Practicing the Business Activity without an environmental Permit.	The first category is 5,000 The second category is 20,000 The third category is 50,000	The violator shall also eliminate the violation, repair the damage and pay compensation.
3	Starting the establishment of the Business Activity without a valid Environmental Permit for Establishment.	The first category is 5,000 The second category is 20,000 The third category is 50,000	The violator shall also eliminate the violation, repair the damage and pay compensation.
4	Staring the operational phase of a Business Activity without a valid Environmental Permit for Operation.	The first category is 10,000 The second category is 20,000 The third category is 50,000	The violator shall also eliminate the violation, repair the damage and pay compensation.
5	Making any amendment or addition to any authorized Business Activity without the approval of the Center.	The first category is 5,000 The second category is 20,000 The third category is 50,000	The violator shall also eliminate the violation, repair the damage and pay compensation.
6	Failure to comply with the requirements of the environmental Permit.	The first category is 1,000 (per condition) The second category is 2,000 (per condition) The third category is 10,000 (per condition)	The violator shall also eliminate the violation, repair the damage and pay compensation.
7	Non-compliance with the implementation of the	The second category is 20,000 (per requirement)	

No.	Violation	Fine/SAR	Notes
	Environmental Impact Assessment Study obligations.	The third category is 50,000 (per requirement)	The violator shall also eliminate the violation, repair the damage and pay compensation.
8	Violation of the controls and requirements for closing the Business Activity as specified by the Center.	The first category is 5,000 (per condition)	The violator shall also eliminate the violation, repair the damage and pay compensation.
		The second category is 20,000 (per condition)	
		The third category is 50,000 (per condition)	
9	The violator shall also eliminate the violation, repair the damage and pay compensation.	The first category is 500 (per requirement)	The violator shall also eliminate the violation, repair the damage and pay compensation.
		The second category is 2,000 (per requirement)	
		The third category is 5,000 (per requirement)	

APPENDIX B

Construction Environment and Sustainability Management Plan (CESMP) Development Guidelines

The CESMP shall include, but not be limited to, the following components:

- Project description including location, scope of work, specific method statements and procedures, environmental permits, and approvals
- Environmental policy, objectives, targets, KPIs, environmental roles and responsibilities, communication, reporting, non-conformances, training and awareness, procurement, subcontractor selection, and management review
- Environmental aspects, impacts and controls for air, surface water and storm water, soil and groundwater, noise and vibration, traffic, water use, material consumption, waste management, carbon emissions, and environmental incidents and emergencies
- Environmental monitoring, inspection, and auditing requirements
- Environment completion process at the demobilisation phase
- Attachments such as legal register, aspects and impacts register, permitting and compliance register, and applicable management system checklists
- The following specific Environment and Sustainability Management Plans (ESMPs) shall be incorporated in the CESMP:

    a) Construction and Demolition (C&D) Waste Management Plan, to ensure that the following is achieved:
        1. A minimum of 75 % of C&D waste (combined) is diverted from landfills for reuse / recycling
        2. A minimum of 60 % of demolition waste is diverted from landfills for reuse / recycling

3. A minimum of 50 % of construction waste is diverted from landfills for reuse / recycling
4. Maximise the recycling / reuse of hazardous waste

The plan shall include the following, as applicable:

- Responsible parties and contact information
- Waste forecasting, monitoring, and reporting
- Waste generation sources and methods for handling, storage, and disposal
- Identify at least five materials (both structural and non-structural) targeted for diversion from landfills and recycling
- Approximate the percentage of the overall project waste that each of these five identified materials represent
- Waste reduction, reuse, and recycling actions, including but not limited to:
    - Purchasing only necessary materials, avoiding double packaging, avoid use of disposable materials, avoid temporary support systems, print double sided and use soft copies
    - Reuse packaging material, reuse metal and wood waste, crush, and reuse reclaimed asphalt and concrete, pour excess concrete into moulds for temporary use (i.e. brick paving, barriers)
    - Dedicated waste management area and trained waste management team, clearly labelled bins for segregation and collection of waste per type
- Provide locations where specific materials will be taken for recycling, and detail how the specific materials are processed
- Include monthly waste diversion, on-site and off-site reuse and recycling, and disposal by waste type / category
- Include receipts, waste manifests, and licences from waste companies as applicable

Handling of hazardous and toxic substances on site shall be organised in accordance with local health, safety, and environmental regulations and shall be carried by NCEC-approved subcontractors.

b) Construction Materials Transportation and Procurement Plan, to ensure that the following is achieved:
   1. 40 % (by cost) of permanently installed top five infrastructure / public realm materials that meet at least one of the following sourcing and extraction requirements:
      i. Extended producer responsibility (EPR), valued at 50 % of cost for purpose of credit calculations
      ii. Material reuse, valued at 200 % of cost for purpose of credit calculations
      iii. Recycled content, valued at 100 % of cost for purpose of credit calculations

iv. USGBC approved program meeting responsible sourcing and extraction criteria
2. 30 % (by cost) of permanently installed infrastructure / public realm materials are sourced from within KSA
3. 10 % (by cost) of permanently installed infrastructure / public realm materials are sourced from at least one of the following:
    i. Reused materials (excluding aggregates)
    ii. Reclaimed timber
    iii. Recycled rubber
    iv. Recycled steel
4. 50 % (by weight) of all aggregates used for permanent installation of infrastructure / public realm are recycled aggregates.

The plan shall include a procurement policy, strategy, and screening process for supplier appointments and product/material selections, including but not limited to the following, as applicable:

- Review the environment and sustainability credentials of suppliers, manufacturers, and specific products/materials
- Prioritise products/materials that are sourced close to site and within KSA
- Prioritise suppliers and manufacturers with documented best environment and sustainability practices, and have validated environmental product declarations

c) Construction Energy and Water Consumption Reduction Plan, to ensure that the following is achieved:
1. Energy. Implement at least three of the following conservation strategies:
    i. Use certified energy efficient generators, office appliances, implement automated lighting controls and install renewable energy sources
    ii. Ensure controlled operation of air conditioning systems
    iii. Ensure that workers live within a 25 km radius of site, transportation requirements are coordinated, and fuel-efficient transportation methods are implemented
    iv. Monitor staff transportation and ensure that efficient transportation methods are implemented
    v. Monitor and record energy usage against a pre-determined baseline, and implement mitigation measures if consumption levels exceed expected range
2. Water. Implement at least three of the following conservation strategies:
    i. Create a water conservation strategy that minimises water wastage through user negligence, inefficient operation, and leakages
    ii. Install water efficient fixtures on faucets, toilets and other as applicable

iii. Use non potable water for construction processes e.g. secondary treatment and recycling of wastewater for irrigation, dust control, and flushing
iv. Limit access to water usage equipment to trained users only
v. Monitor and record water usage against a pre-determined baseline, and implement mitigation measures if consumption levels exceed expected range

d) Noise and Vibration Management Plan, to ensure that the following is achieved:
1. Minimise and control adverse impacts from construction related noise and vibration emissions
2. Planned and effective weekly monitoring of noise and vibration levels at the site boundary and near sensitive receptors, using compliant and approved equipment
3. Where noise generating construction activities are conducted within 100 metres of occupied public sensitive receptors, noise monitoring shall be conducted daily
4. Maintain a daily log of noise observations, including activities near sensitive receptors and mitigation actions if required
5. NCEC noise and vibration limits and thresholds shall be followed, and appropriate mitigation measures implemented (according to accepted industry standards), including but not limited to the following, as required:
   i. Noise attenuation barriers
   ii. Noise attenuation attachments for rock breakers
   iii. Alternative machinery selection (e.g. drum cutter)
   iv. Coordination with external stakeholders (e.g. notifications, agreed working hours)

e) Air Quality Management Plan, to ensure that the following is achieved:
1. Minimise and control adverse impacts from construction related dust emissions.
2. Planned and effective monthly monitoring of air quality (PM10 and PM2.5) at the site boundary and near sensitive receptors, using compliant and approved equipment
3. Where dust generating construction activities are conducted within 100 metres of occupied public sensitive receptors, air quality monitoring shall be conducted daily
4. Maintain a daily log of air quality observations, including meteorological conditions (wind speed, wind direction etc.), dust plumes, construction activities near sensitive receptors, activities near site boundary, and mitigation actions if required
5. NCEC air quality limits and thresholds shall be followed, and appropriate mitigation measures implemented (according to accepted industry standards), including but not limited to the following, as required:

i. Isolate dust producing activities and position stockpiles away from sensitive receptors
ii. Reduce stockpile heights, regularly sprinkle with suitable water and cover with hessian / green mesh
iii. Control vehicle speeds and routes on internal haul roads
iv. Gravel / haul roads: Regularly sprinkle suitable water, maintain through grading of fine material, and establish with reclaimed asphalt pavement and / or aggregate
v. Apply non-hazardous dust control product on haul roads or other applicable dust generating open spaces
vi. Cover truck bins with netting during transportation of material
vii. Sprinkle suitable water during material movement (i.e. loading / offloading)
viii. Fix crushing and screening equipment with sprinkler system, and use water canon / mister
ix. Establish dust screens around dust generating activities and stockpiles
x. Stop dust generating activities during periods of high wind speeds (i.e. 36 km/h)

6. Minimise and control adverse impacts from construction related gas emissions. NCEC air quality limits and thresholds shall be followed, and appropriate mitigation measures implemented (according to accepted industry standards), including but not limited to the following, as required:
   i. Perform onsite measurements of emissions in the air during throughout construction period
   ii. Conduct regular maintenance of plant, equipment, and machinery according to manufacturers' specifications
   iii. Purchase low sulphur fuels

f) Site Inspection and Audit Plan, to ensure that the following is achieved:
   1. Regular site inspections and audits are conducted to identify issues, ensure that corrective and preventative measures are implemented, and continual improvement is promoted.

   The frequency of site inspections and audits shall be based on the duration of the project and specific activities that pose variable environmental risks. After each site inspection / audit a report shall be produced that must include, but not limited to, the following:

   i. Date stamped photographs
   ii. An update on all priority CESMP plans
   iii. Responses to observed deviations from the CESMP and legal requirements with corrective and preventative measures
   iv. Verification that all corrective and preventative measures have been implemented

g) Soil, Erosion and Sedimentation Control Plan
h) Ecology Management Plan
i) Cultural Heritage Management Plan
j) Water and Wastewater Management Plan
k) Hazardous Materials Management Plan
l) Incident and Emergency Management Plan
m) Non-conformance Management Plan
n) Subcontractors Management Plan
o) Traffic and Transport Management Plan
p) Specific Sustainability Management Plans

It is the Contractor's responsibility to ensure compliance with all legal requirements and project deliverables.

APPENDIX C

Environment Monthly Report (EMR) Development Guidelines

The EMR shall include the following information and data as a minimum:

1. Organisations
    a. Contractor Company
    b. CSC Company
    c. PMC Company
    d. EMC Company (if applicable)
    e. Specialist Subcontractor Companies, for example:
        i. Hazardous / Non-hazardous management, sewage waste, water supply, drilling/tunnelling, dewatering, crushing and screening, haulage, monitoring.

2. Overview of Project Schedule
    a. Brief description of project's scope of work (no more than 5 lines / points)
    b. Project start date
    c. Anticipated project end date
    d. Asset-specific construction progress according to broad activity categories, per the tables below (NA where not applicable):

        i. Infrastructure Projects

Activity Description	Completion (%)
OVERALL Construction Completion	?? %

Site Clearance	?? %
Earthworks	?? %
Utilities	?? %
Roadworks	?? %
Landscaping	?? % / NA

ii. Buildings Projects

Activity Description	Completion (%)
OVERALL Construction Completion	?? %
Site Clearance	?? %
Earthworks	?? %
Core & Shell	?? %
Utilities & Fit Out	?? %
Landscaping	?? % / NA

iii. Landscaping Projects

Activity Description	Completion (%)
OVERALL Construction Completion	?? %
Site Clearance	?? %
Earthworks	?? %
Utilities	?? %
Hardscape	?? %
Softscape	?? %

3. Location Maps
    a. Map of project boundary, interfacing projects and sensitive receptors
    b. Map of project facilities (i.e. batching plant, site office, laydown area, material stockpiles)
    c. Map of environmental monitoring locations

4. Environmental Monitoring
    Information and data presented in tables and graphs for air quality, noise and vibration, and excavated and imported material monitoring detailing the following:
    a. Location, frequency, and duration of monitoring conducted during the month
    b. Comparison of results against NCEC limits and baseline data
    c. Description of mitigation measures (implemented and / or planned) to reduce exceedances, if applicable
    d. Monthly and cumulative data (tables and graphs) showing monitoring results from project start date up to and including the reporting month

5. Waste Management

Waste data shall be reported in metric tonnes, and if waste quantities are not physically weighed the density conversion factors (DCFs) detailed in Tables D-1 to D-3 shall be applied to convert m^3 to tonnes. Other applicable volume conversion factors (VCFs) and DCFs are detailed in Table D-4. Data and information related to waste management shall include, but no be limited to the following:

a. Non-hazardous Waste (sent for Disposal)

Waste Type	**Quantity Disposed (tonnes)**	**Number of Trips**	**Distance from Site to Disposal Location (km)**	**Name / Reference of Disposal Location**
Food				
Plastic (General)				
Plastic HDPE				
Paper / Cardboard				
Wood				
Electronic (Non-hazardous)				
Used Tyres				
Geotextile (Non-Woven Polypropylene)				
Vegetation / Green				
Mixed General				
Topsoil				
Wadi Gravel				
Excavated Material				
Broken Concrete				
Concrete Paving / Kerbstones				
Reclaimed Asphalt Pavement (RAP)				
Steel / Metals				
Unsuitable Excavated Material				
Adobe Bricks				
Limestone Blocks				
Glass				
Mixed Demolition				
Unsuitable Dry Slurry Material				
Other				

b. Non-Hazardous Waste (sent for Recycling)

Waste Type	Quantity Recycled Directly Into Project (tonnes)	Quantity Transported to Recycling Facility (tonnes)	Number of Trips	Distance from Site to Recycling Location (km)	Name / Reference of Recycling Location
Plastic (General)					
Plastic HDPE					
Paper / Cardboard					
Wood					
Electronic (Non-hazardous)					
Used Tyres					
Geotextile (Non-Woven Polypropylene)					
Vegetation / Green					
Mixed General					
Topsoil					
Wadi Gravel					
Excavated Material					
Broken Concrete					
Concrete Paving / Kerbstones					
Wet / Excess Concrete					
Reclaimed Asphalt Pavement (RAP)					
Excess Asphalt					
Steel / Metals					
Adobe Bricks					
Limestone Blocks					
Glass / Broken Façade					
Mixed Demolition					
Other					

c. Other Non-Hazardous Waste (Retained on Site for Recycling / Reuse)

Waste Type	Quantity Recycled Directly Into Project (m^3)	Quantity Transported for Site Treatment (m^3)	Number of Trips	Distance to Site Treatment Location (km)	Reference of Site Treatment (e.g. Area 2 Demolition)
Concrete					
Adobe Bricks					
Limestone Blocks					
Other					

d. Solid Hazardous Waste (sent for Disposal)

Waste Type	**Quantity Disposed (tonnes)**	**Number of Trips**	**Distance from Site to Disposal Location (km)**	**Name / Reference of Disposal Location**
Contaminated Soil				
Contaminated Rags / Absorbents				
Empty Metal Drums / Containers				
Empty Plastic Drums / Containers				
Used Batteries				
Used Oil/Fuel Filters				
Electronic (Hazardous)				
Asbestos				
Medical				
Other				

e. Solid Hazardous Waste (sent for Recycling)

Waste Type	**Quantity Recycled (tonnes)**	**Number of Trips**	**Distance from Site to Recycling Location (km)**	**Name / Reference of Recycling Location**
Empty Metal Drums / Containers				
Empty Plastic Drums / Containers				
Used Batteries				
Used Oil/Fuel Filters				
Electronic (Hazardous)				
Other				

f. Liquid Hazardous Waste (sent for Disposal)

Waste Type	**Quantity Disposed (litres)**	**Number of Trips**	**Distance from Site to Disposal Location (km)**	**Name / Reference of Disposal Location**
Used Engine Oil				
Used Transformer Oil				
Oily Water				
Other				

g. Liquid Hazardous Waste (sent for Recycling)

Waste Type	**Quantity Recycled (litres)**	**Number of Trips**	**Distance from Site to Recycling Location (km)**	**Name / Reference of Recycling Location**
Used Engine Oil				

Used Transformer Oil				
Oily Water				
Other				

h. Sewage Waste (sent for Disposal)

Waste Type	Quantity Disposed (m³))	Number of Trips	Distance from Site to Disposal Location (km)	Name / Reference of Disposal Location
Sewage				

i. Other Water Waste (sent for Disposal)

Waste Type	Quantity Disposed (m³)	Number of Trips	Distance from Site to Disposal Location (km)	Name / Reference of Disposal Location
Groundwater / Dewatering				
Tunnelling Water				
Washing / Grey Water				
Other				

j. Other Water Waste (Retained on Site for Recycling / Reuse)

Waste Type	Quantity Recycled Directly Into Project (m³)	Quantity Transported for Site Treatment (m³)	Number of Trips	Distance to Site Treatment Location (km)	Reference of Site Treatment (e.g. Lagoon)
Groundwater / Dewatering					
Tunnelling Water					
Washing / Grey Water					
Other					

A consolidated tracker relating to waste data shall be presented, along with graphs and tables, and shall include cumulative quantities from the project start up to and including the reporting month according to the following criteria:

a. Summary of non-hazardous waste according to type
b. Summary of non-hazardous waste according to use and destination
c. Summary of hazardous waste according to type

d. Summary of hazardous waste according to use and destination
e. Summary of spoil quantities according to type
f. Summary of spoil quantities according to use and destination
g. Summary of wastewater quantities according to type
h. Summary of wastewater quantities according to use and destination
i. Summary of waste diverted from landfill / disposal facility
j. Summary / update on the waste quantities compared to initial waste forecast and progress related to quantities diverted from landfill / disposal location. Waste quantities forecast shall be reviewed and revised if required.

6. Imported / Consumed Materials

a. Water

Water Type	**Quantity Imported / Consumed (m^3)**	**Use**	**Number of Trips**	**Distance from Water Source to Site (km)**	**Name / Reference of Water Source Location**
Potable (Bottles / Drums)					
Potable (Tankered)					
Non-potable					
TSE					
Recycled Groundwater / Dewatering					
Recycled Tunnelling Water					
Recycled Washing / Grey Water					
Other					

b. Hydrocarbons

Hydrocarbon Type	**Consumed (litres)**	**Use**
Diesel		
Biodiesel		
Petrol		
Engine Oil		
Transformer Oil		

Other		

c. Permanent and / or Temporary Works Materials

Material Type	Quantity Imported / Consumed (tonnes)	Use	Number of Trips	Distance from Material Source to Site (km)	Name / Reference of Material Source Location
Fill Material (Recycled by Project/Employer)					
Fill Material (External Source)					
Sub-base (Recycled by Project/Employer)					
Sub-base (External Source)					
Road Base (Recycled by Project/Employer)					
Road Base (External Source)					
Gabbro Aggregate					
Dune Sand					
Washed Sand					
Asphalt (standard)					
Asphalt (with recycled content)					
Limestone Blocks					
Adobe Bricks					
Cement					
Pre-mix Concrete (standard)					
Pre-mix Concrete (with recycled content)					
Curing Compound					
Precast Concrete Manholes*					
Precast Concrete Pipes*					
Concrete Masonry Units*					
Concrete Paving Blocks*					
Concrete Kerbstones*					
Vitrified Clay Piping					
HDPE Piping					
PVC Piping					
GRP Piping					

Material Type	Quantity Imported / Consumed (tonnes)	Use	Number of Trips	Distance from Material Source to Site (km)	Name / Reference of Material Source Location
Steel (Rebar)					
Metal (Copper) Piping					
Metal (Other)					
Cable					
HDPE Sheeting					
Geotextile (Non-Woven Polypropylene)					
Green Mesh					
Hessian					
Timber / Wood (Non-FSC)					
Timber / Wood (FSC certified)					
Paper A4					
Paper A3					
AC Units					
AC Ducting					
Metal Window Frames					
Glass Windows / Facades					
Liquid Petroleum Gas (LPG)					
Oxygen					
Acetylene					
Interior Finishes 1					
Interior Finishes 2					
Interior Finishes 3					
Other					

*Finished concrete material types can be detailed under "Use" if produced by Contractor's own batching plant with pre-mix concrete.

Material consumption data shall be presented with graphs and tables, and shall include cumulative quantities from the project start up to and including the reporting month according to the following criteria:

a. Summary of water quantities according to type
b. Summary of water quantities according to use and source
c. Summary of hydrocarbon quantities according to type
d. Summary of hydrocarbon quantities according to use
d. Summary of permanent and / or temporary works materials according to type
e. Summary of permanent and / or temporary works materials according to use
f. Summary / update on procured materials and forecast related to requirements detailed in the Construction Materials Transportation and Procurement Plan.

A consolidated tracker with comments relating to the sustainable sourcing of materials shall be included, as well supporting documentation for each material, as applicable. For example, if timber is supplied by an FSC certified supplier, then the FSC certificate shall be included, and if premixed concrete includes a percentage of fly ash replacement, then the concrete mix design shall be included as supporting evidence.

7. Environmental Registers
   Environmental registers shall include, but not be limited to the following:
   a. Environmental monitoring
   b. Inspections and audits
   c. Hazardous materials stored on site
   d. Environmental training
   e. Environmental observations (NCRs, issues)
   f. Environmental violations / penalties issued by competent authorities / public complaints
   g. Tree relocation register
   h. Environmental permitting register
   i. Environmental permit compliance tables / mechanisms

Quarterly, or bi-annual Environment and Sustainability Performance Reports (ESPR) shall be compiled by the Contractor and submitted according to the Employer's requirements and specific Project Environmental Permit, to satisfy NCEC requirements.

Table D-1: Density Conversion Factors for Solid Non-hazardous Waste

Waste Type	Multiply m³ Quantity by Factor Below to Obtain Quantity in Metric Tonnes (1 tonne = 1,000 kg)
Food	0.4
Plastic (General)	0.5
Plastic (HDPE)	0.8
Paper / Cardboard	0.4
Wood	0.5
Electronic (Non-hazardous)	0.5
Used Tyres	0.5
Geotextile (Non-Woven Polypropylene)	0.5
Vegetation / Green	0.5
Mixed General	1.2
Topsoil	1.6
Wadi Gravel	1.6
Excavated Material	1.6
Broken Concrete	1.6
Concrete Paving / Kerbstones	1.8
Reclaimed Asphalt Pavement (RAP)	1.6
Steel / Metals	1.6
Unsuitable Excavated Material	1.6
Adobe Bricks	1.6
Glass / Broken Façade	0.8
Mixed Demolition	1.6
Unsuitable Dry Slurry Material	1.6

Other	1 (as applicable)

Table D-2: Density Conversion Factors for Solid Hazardous Waste

Waste Type	Multiply m³ Quantity by Factor Below to Obtain Quantity in Metric Tonnes (1 tonne = 1,000 kg)
Contaminated Soil	1.5
Contaminated Rags / Absorbents	0.5
Empty Metal Drums / Containers	0.6
Empty Plastic Drums / Containers	0.5
Used Batteries	2
Used Oil/Fuel Filters	1.6
Electronic (Hazardous)	0.5
Asbestos	1.6
Medical	0.4
Other	1 (as applicable)

Table D-3: Density Conversion Factors for Imported / Consumed Materials

Material Type	Multiply m³ Quantity by Factor Below to Obtain Quantity in Metric Tonnes (1 tonne = 1,000 kg)
Fill Material (Recycled by Project/Employer)	1.6
Fill Material (External Source)	1.6
Sub-base (Recycled by Project/Employer)	1.7
Sub-base (External Source)	1.7
Road Base (Recycled by Project/Employer)	1.7
Road Base (External Source)	1.7
Gabbro Aggregate	1.7
Dune Sand	1.7
Washed Sand	1.7
Asphalt (standard)	2
Asphalt (with recycled content)	2
Limestone Blocks	2.2
Adobe Bricks	1.6
Cement	NA (kg / tonnes)
Pre-mix Concrete (standard)	2.2
Pre-mix Concrete (with recycled content)	2.2
Curing Compound	1
Precast Concrete Manholes*	2
Precast Concrete Pipes*	2
Concrete Masonry Units*	2
Concrete Paving Blocks*	2
Concrete Kerbstones*	2
Vitrified Clay Piping	2
HDPE Piping	1
PVC Piping	0.8
GRP Piping	0.8

Material Type	Multiply m³ Quantity by Factor Below to Obtain Quantity in Metric Tonnes (1 tonne = 1,000 kg)
Steel (Rebar)	2.2
Metal (Copper) Piping	0.8
Metal (Other)	2.0
Cable	2.0
HDPE Sheeting	0.8
Geotextile (Non-Woven Polypropylene)	0.8
Green Mesh	0.4
Hessian	0.4
Timber / Wood (Non-FSC)	0.5
Timber / Wood (FSC certified)	0.5
Paper	NA (kg / tonnes)
AC Units	1.8
AC Ducting	0.8
Metal Window Frames	0.8
Glass Windows / Facades	2.0
Grease	NA (kg / tonnes)
Liquid Petroleum Gas (LPG)	NA (kg / tonnes)
Oxygen	NA (kg / tonnes)
Acetylene	NA (kg / tonnes)
Interior Finishes 1	1 (as applicable)
Interior Finishes 2	1 (as applicable)
Interior Finishes 3	1 (as applicable)
Other	1 (as applicable)

Table D-4: Volume and Density Conversion Factors for Liquid Materials (reported in m³)

Waste Type	Convert US Gallons (USG) by Factor Below to Obtain Quantity in Litres	Multiply Litres Quantity by Factor Below to Obtain Quantity in m³
Surface / Groundwater	3.785	0.001
Sewage	3.785	0.0008
Diesel	3.785	0.0009
Biodiesel	3.785	0.0009
Petrol	3.785	0.0008
Engine Oil	3.785	0.0009
Transformer Oil	3.785	0.0009

VOLUME 2 – SPECIFICATION

DGCL Sustainability Requirements

DGCL Sustainability Scope of Services for Contractors

1. Contractor Core Sustainability Roles and Responsibilities

- Implement all plans and construct the Project in accordance with the sustainability design intent
- Prepare a Construction Environmental and Sustainability Management Plan (CESMP [Refer to the Environmental Scope of Services for Contractors] and submit to the Construction Supervision Consultant (CSC), within 28 days of contract award. Obtain approval of the CESMP from the CSC, PMC and DGCL.
- Appoint at least one full time, site-based Sustainability Professional that will be the focal point between the Contractor and CSC, PMC, DGCL and sustainability certification entities for all sustainability matters (refer to Section 3), from the commencement date until the Project's sustainability certifications are obtained and project is closed out.
- Administer and pay all associated construction review fees and certification fees for the certifying organizations (GBCI, Sustainable Building and other relevant entities)
- Develop all construction sustainability certification submittals. Solicit, collect, and maintain required documentation from the Sub-Contractors to prepare the sustainability certification submittals.
- Review and update the certification submittals and monthly report on the sustainability progress throughout the construction period.
- Implement commissioning requirements of the Project or perform the required commissioning tasks, depending on the Project pursued commissioning criteria
- Manage and administer visits, audits, and inspections conducted on-site by certification bodies and authorities related to environment and sustainability, and timely respond to the non-compliance issues raised during visits, audits, and inspections
- Obtain the final sustainability certifications with the Project intended rating no later than four (4) months of practical completion
- Prepare and maintain the sustainability construction material review tracker in accordance with DGCL's requirements, and report the progress monthly
- Collect the construction carbon (greenhouse gases) emissions data from construction activities and report the progress monthly
- Implement indoor air quality testing in accordance with LEED and Mostadam requirements
- Compile Sustainability Monthly Reports, and submit to the CSC for review and approval, per DGCL requirements. Submit Sustainability Monthly Reports in DGCL's PMIS
- Communication and coordination with project stakeholders as required.

2. Sustainability Specifications

DGCL's sustainability development guidelines reflect the sustainable design, construction, and operation requirements to achieve DGCL's Sustainability KPIs and the following certifications, as required for the specific project:

- LEED for Cities & Communities Gold certification (aspirational target of Platinum)
- Mostadam for Communities Gold certification
- SITES Rating System for Sustainable Land Design and Development
- LEED Building Design and Construction
- Mostadam for Residential Buildings Design and Construction
- Mostadam for Commercial Buildings Design and Construction
- Parksmart Gold certification for parking structures

The Contractor is responsible for implementing all requirements during construction to meet and achieve the project's sustainability objectives and targets.

It is the Contractor's responsibility to ensure compliance with the statutory environmental requirements of the Kingdom of Saudi Arabia (KSA). [Refer to the Environmental Scope of Services for Contractors]

Based on the Diriyah Gate Development construction activities, associated environmental aspects and potential impacts, each project shall prepare a project-specific Construction Environment and Sustainability Management Plan (CESMP) to control activities, monitor aspects and minimise environmental impacts. The CESMP shall be compiled by the Contractor's suitably qualified Environment and Sustainability Professional and submitted to the CSC and PMC for approval. The CESMP shall be revised by the Contractor periodically to remain applicable to the project's scope, activities and updated regulatory conditions as required.

2.1 Pre-Construction

Prior to commencement of construction activities, the Contractor shall attend the environment and sustainability induction, after which reference documentation shall be provided for the Contractor to fulfil its obligations [Refer to the Environmental Scope of Services for Contractors].

2.2 Construction Environment and Sustainability Management Plan (CESMP)

The CESMP serves as the primary reference document to ensure compliance with all legal requirements, deliverables, implementation of environmental and sustainability best practices, and details the project's Environmental Management System (EMS), which shall be aligned with ISO 14001 (2015) requirements. Additional CESMP details and requirements are included under section 2.5 (CESMP Details) [Refer to the Environmental Scope of Services for Contractors]

2.3 Sustainability Monthly Reporting

The Contractor shall comply with DGCL's sustainability reporting requirements and processes, which may be revised by DGCL as required, to fulfil updated regulatory conditions and / or sustainability best practices.

The monthly report shall be signed / endorsed by the Contractor's appointed responsible person (i.e. Project Manager), and submitted to the CSC and PMC for review and approval. Monthly reports shall be submitted to DGCL on the 25th of each month.

3. **Sustainability Team**

Contractors shall ensure that adequate resources (in terms of manpower, equipment, or finance) are provided for effective implementation of the CESMP throughout the project's construction phase.

Contractors shall appoint at least one full time, site-based Sustainability Professional that will be the focal point between the Contractor and project entities for all sustainability matters. The number and seniority of required Sustainability Professionals shall be aligned with project-specific deliverables, project scale, and complexity to ensure compliance and delivery of the project's objectives and targets.

Contractors shall ensure that the CV of the proposed Sustainability Professional is submitted to the CSC, PMC and DGCL for review and approval prior to candidate hiring.

As part of monitoring and review by the PMC, if it is observed that Sustainability Professionals are not competent to perform their duties, a recommendation will be made to replace said staff. Contractors shall provide the PMC with advanced notification in writing in case of any replacement/resignation of the Sustainability Professional.

Table 3.1 below should be used as a guideline to determine the specific title and seniority of the Contractor's Sustainability Professionals. [Refer to the Environmental Scope of Services for Contractors for a definition of the general key positions and staff responsibilities]

Table 3.1: Guideline Criteria for Contractor's Sustainability Professionals

Title	Total Experience	Experience in similar Role (Project Scale / Complexity)	Required Qualifications	Preferred Qualifications / Experience
Sustainability Manager	More than 10 Years	8 Years	BSc. In Environmental Science / Engineering or similar	Postgraduate qualification in related discipline
			GCC Experience	Certification with a recognised professional body.
			LEED AP and/or Mostadam AP, and SITES AP	More than 5 Years GCC Experience

Sustainability Specialist	7 Years	5 Years	BSc. In Environmental Science / Engineering or similar	Postgraduate qualification in related discipline
			GCC Experience	Certification with Sustainability Ratings Scheme (LEED AP and/or Mostadam AP, and SITES AP)
Sustainability Officer	4 Years	2 Years	BSc. In Environmental Science / Engineering or similar	GCC Experience.
				Certification with Sustainability Ratings Scheme (LEED AP and/or Mostadam AP, and SITES AP)

2.5 CESMP Details

The CESMP shall include, but not be limited to, the following components:

- Project description including location, scope of work, specific method statements and procedures, environmental permits, and approvals
- Environmental policy, objectives, targets, KPIs, environmental roles and responsibilities, communication, reporting, non-conformances, training and awareness, procurement, subcontractor selection, and management review
- Environmental aspects, impacts and controls for air, surface water and storm water, soil and groundwater, noise and vibration, traffic, water use, material consumption, waste management, carbon emissions, and environmental incidents and emergencies
- Environmental monitoring, inspection, and auditing requirements
- Environment completion process at the demobilisation phase
- Attachments such as legal register, aspects and impacts register, permitting and compliance register, and applicable management system checklists
- The following specific Environment and Sustainability Management Plans (ESMPs) shall be incorporated in the CESMP:

a. **Construction and Demolition (C&D) Waste Management Plan,** to ensure that the following is achieved:

1. A minimum of 75 % of C&D waste (combined) is diverted from landfills for reuse / recycling
2. A minimum of 60 % of demolition waste is diverted from landfills for reuse / recycling
3. A minimum of 50 % of construction waste is diverted from landfills for reuse / recycling
4. Maximise the recycling / reuse of hazardous waste

The plan shall include the following, as applicable:

- Responsible parties and contact information
- Waste forecasting, monitoring, and reporting
- Waste generation sources and methods for handling, storage, and disposal
- Identify at least five materials (both structural and non-structural) targeted for diversion from landfills and recycling
- Approximate the percentage of the overall project waste that each of these five identified materials represent
- Waste reduction, reuse, and recycling actions, including but not limited to:
    - Purchasing only necessary materials, avoiding double packaging, avoid use of disposable materials, avoid temporary support systems, print double sided and use soft copies
    - Reuse packaging material, reuse metal and wood waste, crush, and reuse reclaimed asphalt and concrete, pour excess concrete into moulds for temporary use (i.e. brick paving, barriers)
    - Dedicated waste management area and trained waste management team, clearly labelled bins for segregation and collection of waste per type
- Provide locations where specific materials will be taken for recycling, and detail how the specific materials are processed
- Include monthly waste diversion, on-site and off-site reuse and recycling, and disposal by waste type / category
- Include receipts, waste manifests, and licences from waste companies as applicable

Handling of hazardous and toxic substances on site shall be organised in accordance with local health, safety, and environmental regulations and shall be carried by NCEC-approved subcontractors.

b. **Construction Materials Transportation and Procurement Plan**, to ensure that the following is achieved:

1. 40 % (by cost) of permanently installed top five infrastructure / public realm materials that meet at least one of the following sourcing and extraction requirements:
    i. Extended producer responsibility (EPR), valued at 50 % of cost for purpose of credit calculations
    ii. Material reuse, valued at 200 % of cost for purpose of credit calculations

iii. Recycled content, valued at 100 % of cost for purpose of credit calculations
iv. USGBC approved program meeting responsible sourcing and extraction criteria

2. 30 % (by cost) of permanently installed infrastructure / public realm materials are sourced from within KSA
3. 10 % (by cost) of permanently installed infrastructure / public realm materials are sourced from at least one of the following:
   i. Reused materials (excluding aggregates)
   ii. Reclaimed timber
   iii. Recycled rubber
   iv. Recycled steel
4. 50 % (by weight) of all aggregates used for permanent installation of infrastructure / public realm are recycled aggregates.

The plan shall include a procurement policy, strategy, and screening process for supplier appointments and product/material selections, including but not limited to the following, as applicable:

- Review the environment and sustainability credentials of suppliers, manufacturers, and specific products/materials
- Prioritise products/materials that are sourced close to site and within KSA
- Prioritise suppliers and manufacturers with documented best environment and sustainability practices, and have validated environmental product declarations

c. **Construction Energy and Water Consumption Reduction Plan,** to ensure that the following is achieved:

1. Energy. Implement at least three of the following conservation strategies:
   i. Use certified energy efficient generators, office appliances, implement automated lighting controls and install renewable energy sources
   ii. Ensure controlled operation of air conditioning systems
   iii. Ensure that workers live within a 25 km radius of site, transportation requirements are coordinated, and fuel-efficient transportation methods are implemented
   iv. Monitor staff transportation and ensure that efficient transportation methods are implemented
   v. Monitor and record energy usage against a pre-determined baseline, and implement mitigation measures if consumption levels exceed expected range

2. Water. Implement at least three of the following conservation strategies:
    i. Create a water conservation strategy that minimises water wastage through user negligence, inefficient operation, and leakages
    ii. Install water efficient fixtures on faucets, toilets and other as applicable
    iii. Use non potable water for construction processes e.g. secondary treatment and recycling of wastewater for irrigation, dust control, and flushing
    iv. Limit access to water usage equipment to trained users only
    v. Monitor and record water usage against a pre-determined baseline, and implement mitigation measures if consumption levels exceed expected range

d. **Noise and Vibration Management Plan,** to ensure that the following is achieved:

1. Minimise and control adverse impacts from construction related noise and vibration emissions
2. Planned and effective weekly monitoring of noise and vibration levels at the site boundary and near sensitive receptors, using compliant and approved equipment
3. Where noise generating construction activities are conducted within 100 metres of occupied public sensitive receptors, noise monitoring shall be conducted daily
4. Maintain a daily log of noise observations, including activities near sensitive receptors and mitigation actions if required
5. NCEC noise and vibration limits and thresholds shall be followed, and appropriate mitigation measures implemented (according to accepted industry standards), including but not limited to the following, as required:
    i. Noise attenuation barriers
    ii. Noise attenuation attachments for rock breakers
    iii. Alternative machinery selection (e.g. drum cutter)
    iv. Coordination with external stakeholders (e.g. notifications, agreed working hours)

e. **Air Quality Management Plan**, to ensure that the following is achieved:

1. Minimise and control adverse impacts from construction related dust emissions.

2. Planned and effective monthly monitoring of air quality (PM10 and PM2.5) at the site boundary and near sensitive receptors, using compliant and approved equipment
3. Where dust generating construction activities are conducted within 100 metres of occupied public sensitive receptors, air quality monitoring shall be conducted daily
4. Maintain a daily log of air quality observations, including meteorological conditions (wind speed, wind direction etc.), dust plumes, construction activities near sensitive receptors, activities near site boundary, and mitigation actions if required
5. NCEC air quality limits and thresholds shall be followed, and appropriate mitigation measures implemented (according to accepted industry standards), including but not limited to the following, as required:
    i. Isolate dust producing activities and position stockpiles away from sensitive receptors
    ii. Reduce stockpile heights, regularly sprinkle with suitable water and cover with hessian / green mesh
    iii. Control vehicle speeds and routes on internal haul roads
    iv. Gravel / haul roads: Regularly sprinkle suitable water, maintain through grading of fine material, and establish with reclaimed asphalt pavement and / or aggregate
    v. Apply non-hazardous dust control product on haul roads or other applicable dust generating open spaces
    vi. Cover truck bins with netting during transportation of material
    vii. Sprinkle suitable water during material movement (i.e. loading / offloading)
    viii. Fix crushing and screening equipment with sprinkler system, and use water canon / mister
    ix. Establish dust screens around dust generating activities and stockpiles
    x. Stop dust generating activities during periods of high wind speeds (i.e. 36 km/h)
6. Minimise and control adverse impacts from construction related gas emissions. NCEC air quality limits and thresholds shall be followed, and appropriate mitigation measures implemented (according to accepted industry standards), including but not limited to the following, as required:
    i. Perform onsite measurements of emissions in the air during throughout construction period

ii. Conduct regular maintenance of plant, equipment, and machinery according to manufacturers' specifications
iii. Purchase low sulphur fuels

f. **Site Inspection and Audit Plan**, to ensure that the following is achieved:

1. Regular site inspections and audits are conducted to identify issues, ensure that corrective and preventative measures are implemented, and continual improvement is promoted.

The frequency of site inspections and audits shall be based on the duration of the project and specific activities that pose variable environmental risks. After each site inspection / audit a report shall be produced that must include, but not limited to, the following:

i. Date stamped photographs
ii. An update on all priority CESMP plans
iii. Responses to observed deviations from the CESMP and legal requirements with corrective and preventative measures
iv. Verification that all corrective and preventative measures have been implemented

g. Soil, Erosion and Sedimentation Control Plan
h. Ecology Management Plan
i. Cultural Heritage Management Plan
j. Water and Wastewater Management Plan
k. Hazardous Materials Management Plan
l. Incident and Emergency Management Plan
m. Non-conformance Management Plan
n. Subcontractors Management Plan
o. Traffic and Transport Management Plan
p. Specific Sustainability Management Plans

It is the Contractor's responsibility to ensure compliance with all legal requirements and project deliverables.